



ITAI 1378 • COMPUTER VISION — MIDTERM PROPOSAL

PPE Compliance Checker

Automated detection of hard hats, hi-vis vests, and safety goggles — flagging every worker as compliant or non-compliant.

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Project Tier: Tier 2 • V1 model trained (mAP@50 0.61)

The Problem

Preventable injuries + compliance liability on the job site

Workers who skip required PPE face some of the most common — and most preventable — worksite injuries: head trauma, struck-by incidents, and eye injuries.

Manual spot-checks don't scale. One safety officer cannot watch every worker continuously, and violations are logged inconsistently, if at all.

The result is both a safety gap and an OSHA-compliance liability.

1

Who cares

Safety officers / EHS managers, superintendents, construction & manufacturing firms, insurers

2

Why it matters

Head, eye & struck-by hazards are leading, largely preventable injuries

3

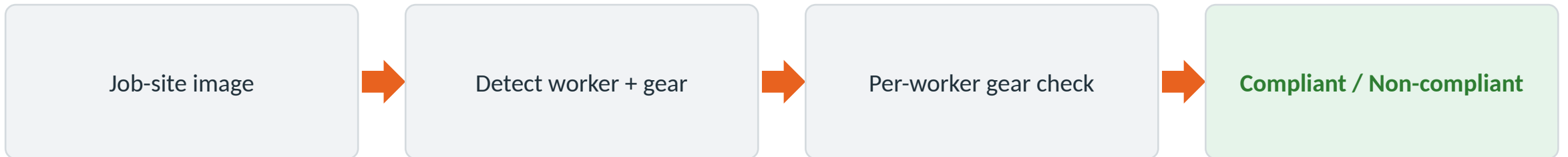
Cost of failure

Direct OSHA citations + injury and liability cost

The Solution

One sentence, one model

An image of a work area goes into a fine-tuned detector that locates each worker and each PPE item, then labels every person “Compliant” or “Non-compliant.”



V2

Capstone (V2): extend to a live or recorded video stream — track each worker across frames and log compliance events with timestamps, making it a deployable, portfolio-ready safety-monitoring tool.

Technical Approach

Object detection, done with the fastest reliable tool

1

TECHNIQUE

Multi-class object detection

Locate + identify people and several gear types in one pass.

2

MODEL

YOLO11 (Ultralytics), fine-tuned

Best accuracy-per-training-hour; trains on free Colab GPUs.

3

FRAMEWORK

PyTorch + Ultralytics

Standard, free, Colab-ready; built-in tracking for V2 video.

4

COMPLIANCE LOGIC

Rule-based, per worker

Associate gear to nearest person; required = hat + vest + goggles.

Why YOLO11: top accuracy-per-training-hour for multi-class detection on a modest budget, and its inference speed is what carries the same model into the real-time V2 capstone without a rewrite.

Data Plan

Public, already-labeled datasets — no manual collection

Roboflow Universe — PPE

Thousands of labeled images (a common set: ~19K / 3.6K / 1.9K)
person, hardhat, vest, goggles + NO- variants

SH17

8,099 images · ~75,994 instances
17 PPE + body-part classes (V2 scale-up)

Ultralytics Construction-PPE ← used for V1

1,416 images (1,132 train / 143 val / 141 test)
11 classes incl. helmet, vest, goggles, no_helmet, no_goggle

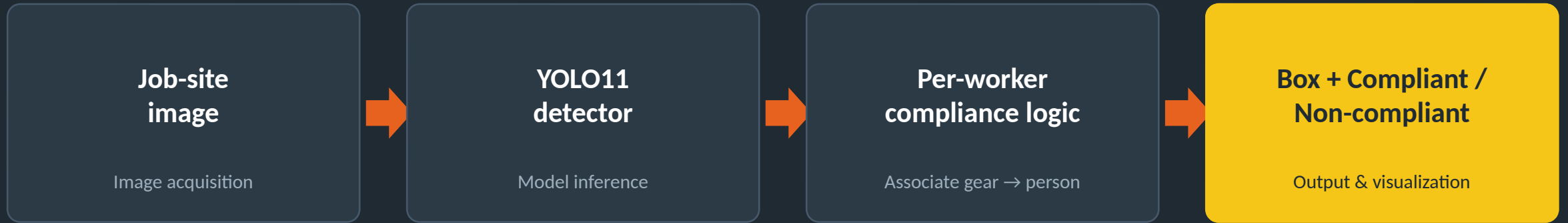
V1 uses Construction-PPE

- 1,416 images total
- 1,132 train / 143 val / 141 test
- Auto-downloads (178 MB), zero setup
- Standardize 640×640, light augmentation

Size: ample for a first fine-tune; scale to SH17 (8,099) for V2.

System Diagram

Input → inference → compliance decision



V2

Capstone: video stream → track workers across frames (ByteTrack) → timestamped compliance log.

Success Metrics

Targets vs. what V1 actually hit

≥ 0.85

mAP@50 (target)

Detection quality across all classes

0.61

mAP@50 (V1 actual)

40 epochs, 1,416 imgs, held-out val

0.30

mAP@50-95 (V1 actual)

Stricter localization metric

Supporting checks

- Precision 0.66 · Recall 0.59 (V1). Model is strong on common gear, weak on rare violation classes (no_goggle, no_boots).
- Latency target < 1 s/image (V1); ≥ 15 FPS for the V2 video capstone.
- 0.61 on a 1,416-image set is an honest baseline — the fix is more data (scale to SH17), not a fancier metric.

V1 result YOLO11s · 40 epochs · Tesla T4 · mAP@50 0.61 · mAP@50-95 0.30 · P 0.66 · R 0.59 (see Appendix for curves + confusion matrix)

Week-by-Week Plan

Six weeks, one milestone each

- | | | |
|---|--|--|
| 1 | Set up Colab; download + merge PPE datasets; explore classes | Dataset ready in YOLO format |
| 2 | Fine-tune YOLO11 baseline on PPE classes | Model trains, first mAP number |
| 3 | Add per-worker compliance logic; tune data | End-to-end compliant/non-compliant on images |
| 4 | Evaluate on held-out test set; improve weak classes | Metrics hit target (≥ 0.85 mAP@50) |
| 5 | Build demo; begin V2 video tracking (stretch) | Demo ready |
| 6 | Final testing, documentation, slides | Everything done → present |

Challenges & Backup Plans

Every risk has a Plan B

CHALLENGE

Rare violation classes (no_goggle, no_boots) score low — confirmed in V1

Small dataset caps accuracy (mAP@50 0.61)

Class imbalance

V2 video more complex than time allows

PLAN B

More data for those classes: scale to SH17 (8,099 imgs) or oversample violations

Larger dataset + larger YOLO11 variant (s→m) + more epochs

Weighted loss, targeted augmentation, oversampling

Keep V2 as capstone/stretch; V1 images is the graded deliverable

Resources Needed

Everything free-tier — total cost ≈ \$0

1

Compute

Google Colab (free T4 GPU) — V1 trained in ~17 min *Colab Pro ~\$10/mo optional*

2

Frameworks

PyTorch · Ultralytics (YOLO11) · OpenCV *\$0 — open source*

3

Data & tools

Construction-PPE · SH17 · Roboflow (free) *\$0*

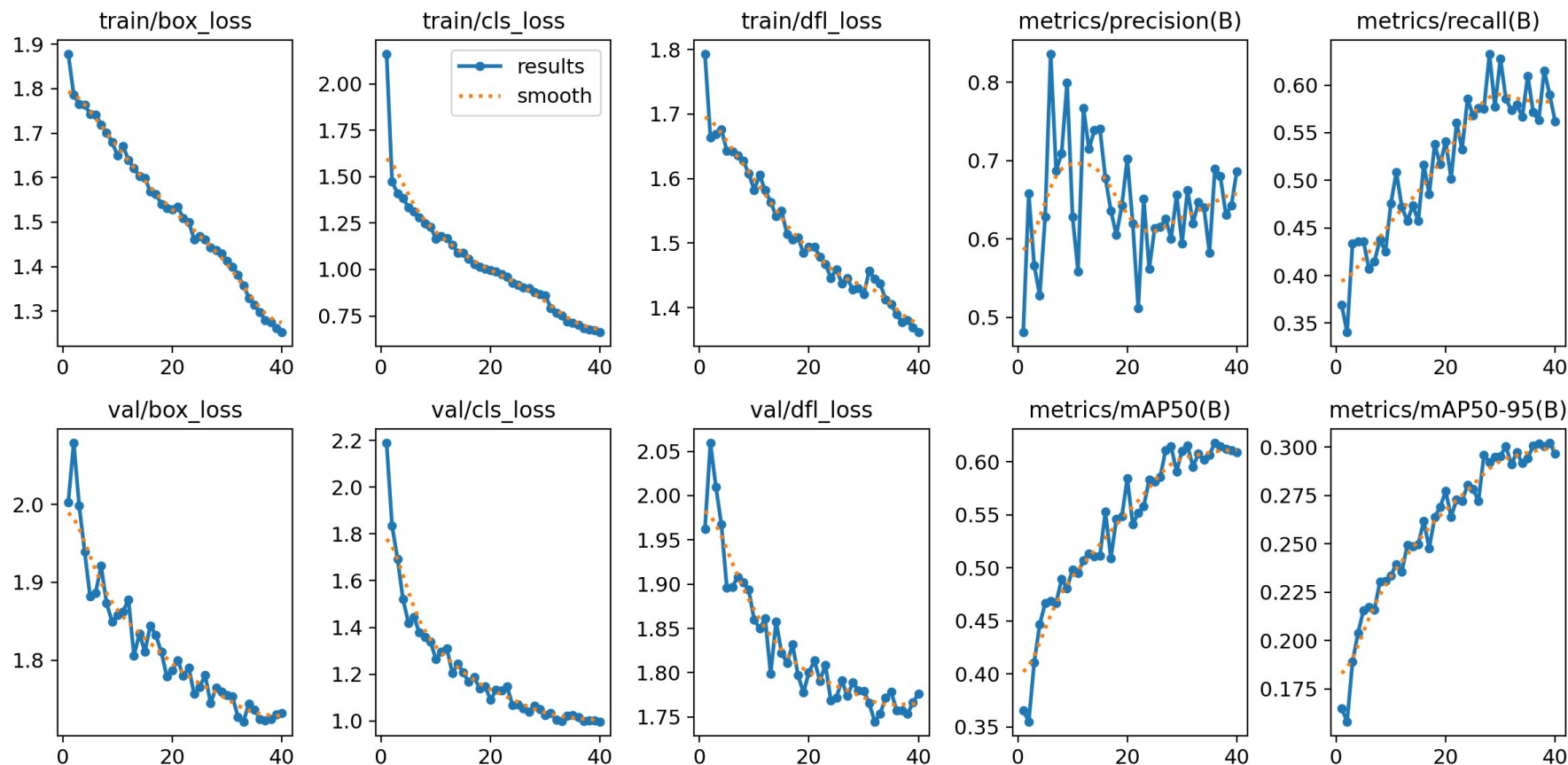
Portfolio payoff

V1 = trained model, mAP@50 0.61 (done).

V2 = real-time video + timestamped logging:
a deployable safety tool for your portfolio.

Appendix — V1 Training Results

YOLO11s • 40 epochs • Tesla T4



Read

- Losses fall smoothly
- mAP climbs, plateaus ~epoch 35–40
- No overfitting or instability
- **mAP@50 → 0.61**

Appendix — Per-Class Results

Where the model is strong vs. weak

Per-class mAP@50

Person	0.533	none	0.216
vest	0.532	no_helmet	0.155
boots	0.450	no_boots	0.085
helmet	0.427	no_goggle	0.077
gloves	0.392	no_gloves	0.077
goggles	0.380		

Insight: strong on common gear (helmet, vest, boots, Person); weak on the rare no_* violation classes the compliance rule depends on. Fix = more violation examples (SH17 / oversampling).

